Green University of Bangladesh

Department of Computer Science and Engineering (CSE)

Faculty of Sciences and Engineering

Semester: (Summer,Year:2026),B.Sc.in CSE (Day)

LAB REPORT NO - 01

Course Title: Computer Networking Lab

Course Code: CSE_318 Section: 241-D2

Lab Experiment Name: SMTP

Student Details

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Lab Report Status	
Marks: …………………………………	Signature:.....................
Comments:...	Date:..............................

1. Objective

- To design and simulate the Green University of Bangladesh (GUB) City Campus network, connecting Building A and Building B through a router.
- To assign IP addresses to all end devices in five departments: CSE, EEE, Textile (Building A) and GBS, Law (Building B).
- To install and configure a dedicated DNS Server and a dedicated Mail Server in each building.
- To configure a DNS A record so each building's mail domain resolves to its Mail Server.
- To configure SMTP/POP3 mail services and create user e-mail accounts for every department.
- To test and verify connectivity and e-mail delivery between Building A and Building B using Cisco Packet Tracer, and to analyse any errors encountered.

2. Theory

2.1 DNS Server

The Domain Name System (DNS) resolves human-readable domain names into IP addresses. In this network, each building has its own DNS Server that stores an A (Address) record mapping the building's mail domain (e.g., mail_a.green.edu.bd) to the IP address of that building's Mail Server.

2.2 Mail Server (SMTP/POP3)

The Mail Server handles sending and receiving of e-mail using SMTP (Simple Mail Transfer Protocol) for outgoing mail and POP3 (Post Office Protocol v3) for incoming mail retrieval. Each Mail Server is configured with a Domain Name and holds a user account for every staff member it serves.

2.3 Router and Switch

Switch2 and Switch3 operate at Layer 2, connecting all devices within their own building into a single LAN. Router2 (Cisco 2811) operates at Layer 3 and connects the two buildings, routing traffic — including e-mail, DNS queries, and ICMP — between the two subnets.

3. Devices and Software Required

- Cisco Packet Tracer (simulation software)
- 1 Router – Router2 (model 2811)
- 2 Switches – Switch2 (Building A), Switch3 (Building B), model 2960-24TT
- 2 DNS Servers – DNS Server A (Building A), DNS Server B (Building B)
- 2 Mail Servers – Mail Server A (Building A), Mail Server B (Building B)
- End devices: 4 PCs and 2 Laptops as listed in the device table below
- Copper straight-through cables for all device-switch and switch-router links

4. Network Topology

The complete City Campus network was built in Cisco Packet Tracer as shown in Figure 1. Building A contains CSE, EEE, and Textile department devices connected to Switch2, along with DNS Server A and Mail Server A. Building B contains GBS and Law department devices connected to Switch3, along with DNS Server B and Mail Server B. Router2 connects the two buildings, with its Building A-side interface addressed 172.134.1.6 and its Building B-side interface addressed 172.135.1.6.

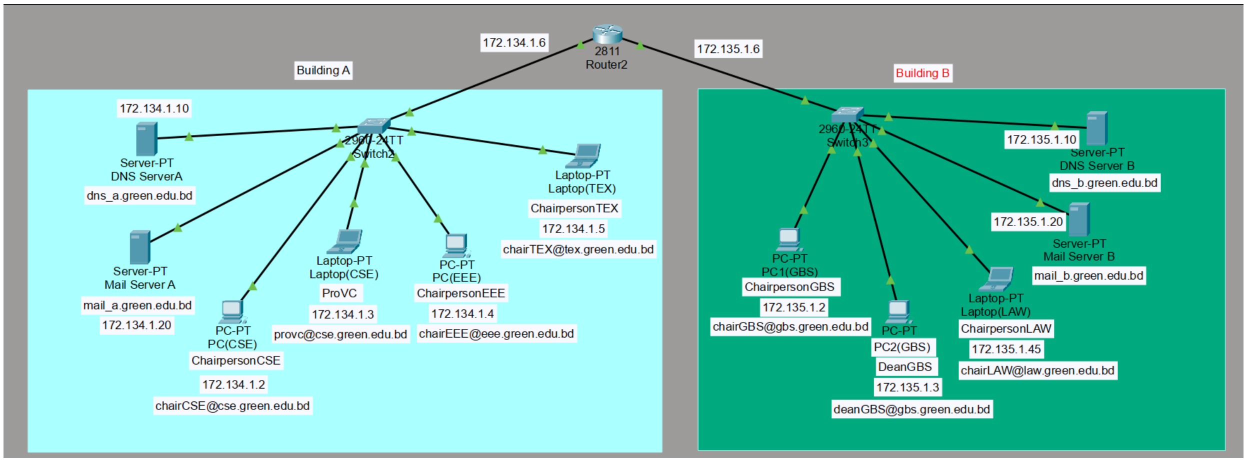

Figure 1: Logical topology of the GUB City Campus network implemented in Cisco Packet Tracer

5. IP Addressing Table

5.1 Building A – Network 172.134.1.0/24

Device Name	Device Type	IP Address	Subnet Mask	Default Gateway
ChairpersonCSE	PC (CSE)	172.134.1.2	255.255.255.0	172.134.1.6
ProVC	Laptop (CSE)	172.134.1.3	255.255.255.0	172.134.1.6
ChairpersonEEE	PC (EEE)	172.134.1.4	255.255.255.0	172.134.1.6
ChairpersonTEX	Laptop (Textile)	172.134.1.5	255.255.255.0	172.134.1.6
DNS Server A	Server	172.134.1.10	255.255.255.0	172.134.1.6
Mail Server A	Server	172.134.1.20	255.255.255.0	172.134.1.6

5.2 Building B – Network 172.135.1.0/24

Device Name	Device Type	IP Address	Subnet Mask	Default Gateway
ChairpersonGBS	PC1 (GBS)	172.135.1.2	255.255.255.0	172.135.1.6
DeanGBS	PC2 (GBS)	172.135.1.3	255.255.255.0	172.135.1.6
ChairpersonLAW	Laptop (Law)	172.135.1.45	255.255.255.0	172.135.1.6
DNS Server B	Server	172.135.1.10	255.255.255.0	172.135.1.6
Mail Server B	Server	172.135.1.20	255.255.255.0	172.135.1.6

5.3 Router Interfaces (Router2 – 2811)

Interface	Connected To	IP Address	Subnet Mask
FastEthernet0/0	Switch2 (Building A)	172.134.1.6	255.255.255.0
FastEthernet0/1	Switch3 (Building B)	172.135.1.6	255.255.255.0

6. DNS Server Configuration

The DNS service on each building's DNS Server was configured with a single A record, resolving that building's mail domain to its own Mail Server IP address, as shown in Figure 2 for DNS Server B.

DNS Server	Record Type	Domain Name	Resolved IP Address
DNS Server A (172.134.1.10)	A Record	mail_a.green.edu.bd	172.134.1.20
DNS Server B (172.135.1.10)	A Record	mail_b.green.edu.bd	172.135.1.20

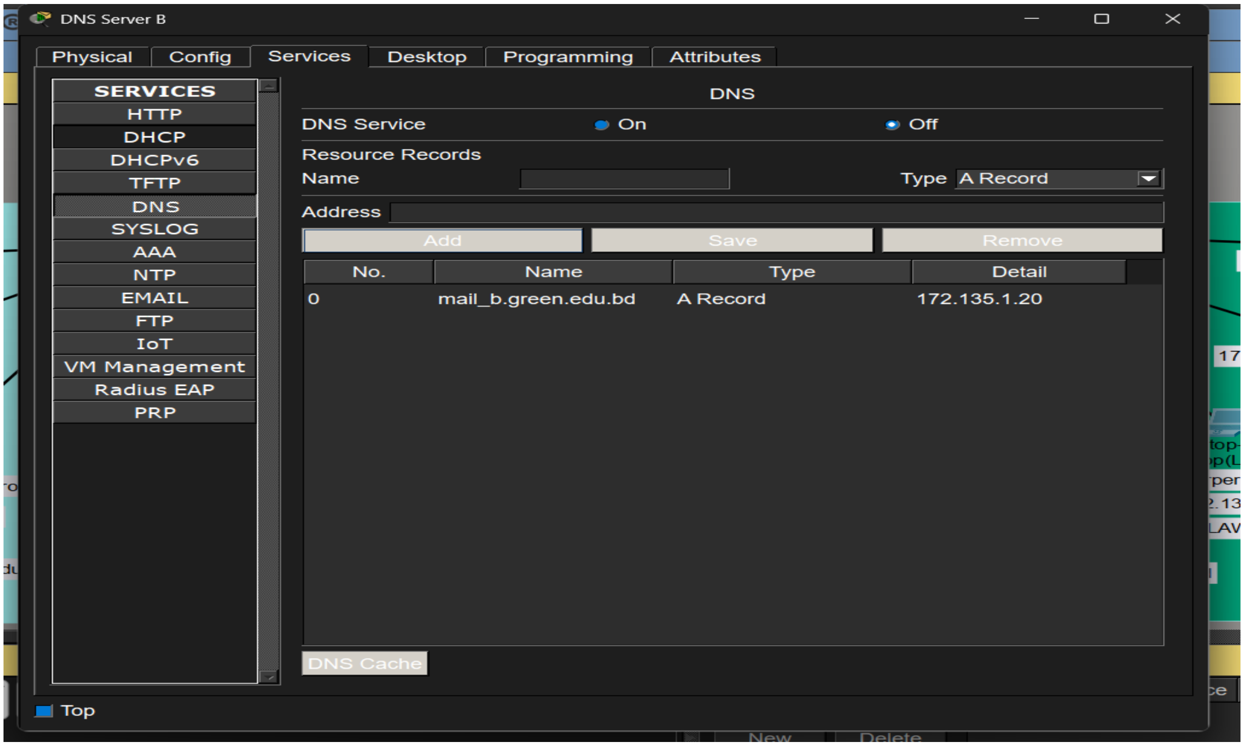

Figure 2: DNS Server B configuration showing the A record for mail_b.green.edu.bd

Observation: at the time of this screenshot, the DNS Service radio button on DNS Server B is still set to Off. For domain-name resolution to work during the e-mail test, this must be switched to On (the same check applies to DNS Server A).

9. Mail Server Configuration

The EMAIL service (SMTP + POP3) was turned on for each building's Mail Server, a Domain Name was registered, and a user account was created for every staff member, as shown in Figures 3 and 4.

9.1 Mail Server A – 172.134.1.20

Domain Name Registered	User Accounts Created
mail_a.green.edu.bd	chircse, provc, chaireee, chairtex

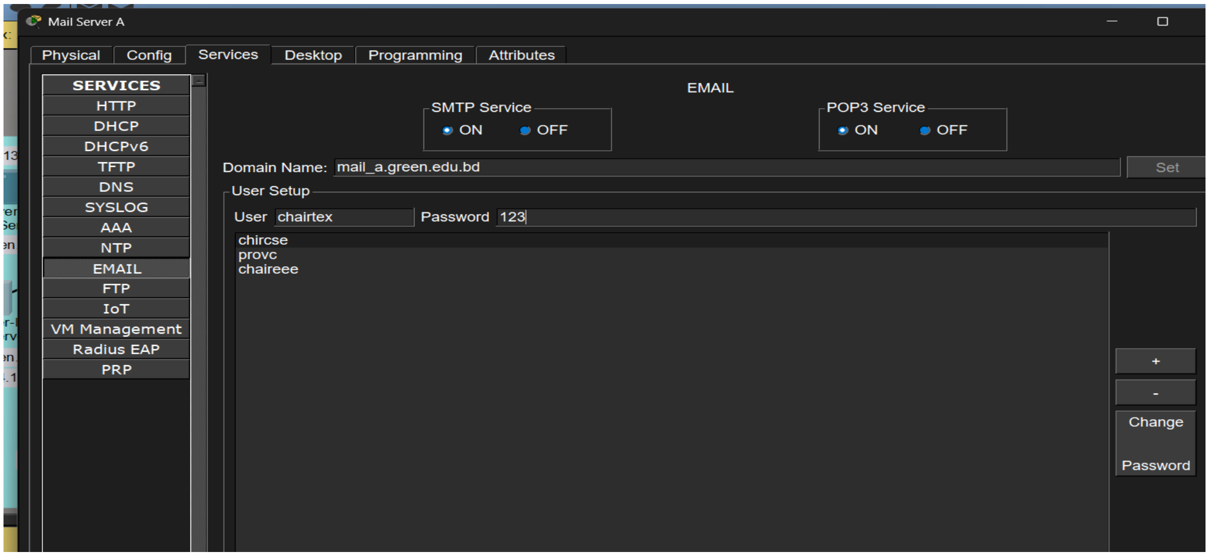

Figure 3: Mail Server A – EMAIL service (SMTP/POP3 ON) with Domain Name mail_a.green.edu.bd and registered user accounts

9.2 Mail Server B – 172.135.1.20

Domain Name Registered	User Accounts Created
mail_b.green.edu.bd	chairgbs, deangbs, chairlaw

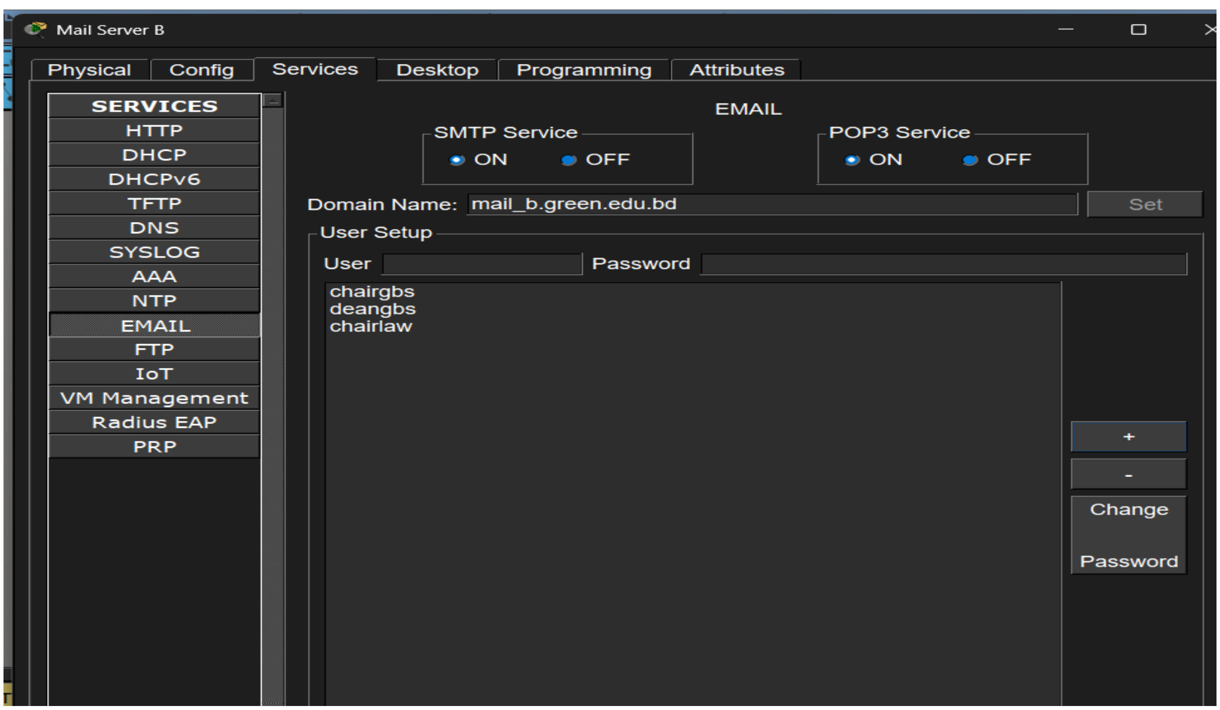

Figure 4: Mail Server B – EMAIL service (SMTP/POP3 ON) with Domain Name mail_b.green.edu.bd and registered user accounts

9.3 E-mail Address List (configured on end devices)

User	E-mail Address (as configured on the device)
ChairpersonCSE	chaircse@cse.green.edu.bd
ProVC	provc@cse.green.edu.bd
ChairpersonEEE	chaireee@eee.green.edu.bd
ChairpersonTEX	chairtex@tex.green.edu.bd
ChairpersonGBS	chairgbs@gbs.green.edu.bd
DeanGBS	deangbs@gbs.green.edu.bd
ChairpersonLAW	chairlaw@law.green.edu.bd

10. End Device Configuration

1. On every PC/Laptop, open Desktop → IP Configuration and set the Static IP Address, Subnet Mask, and Default Gateway (172.134.1.6 for Building A, 172.135.1.6 for Building B) as listed in the IP Addressing Table.
2. Open Desktop → Configure Mail and enter Your Name, Email Address, Incoming/Outgoing Mail Server, User Name and Password, as shown in Figure 5 for PC(CSE).
3. Click Save to store the configuration for that device.

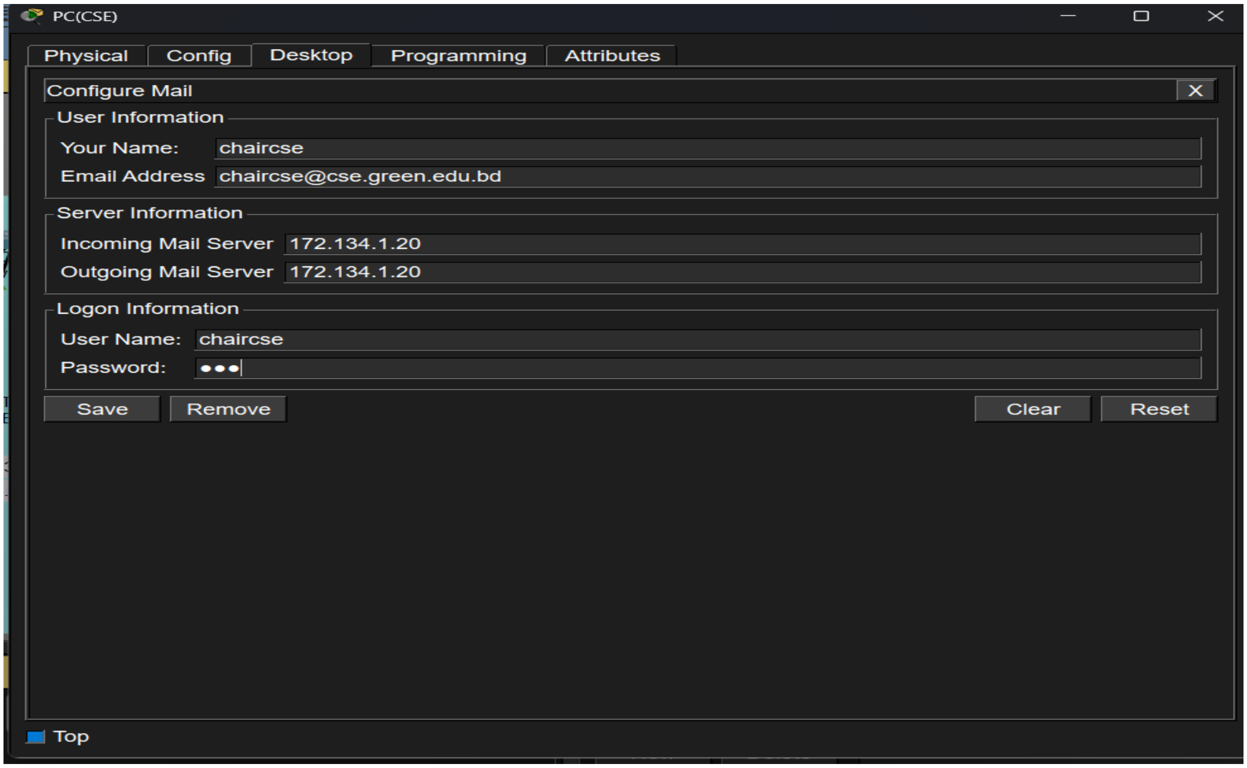

Figure 5: Mail client configuration on PC(CSE) – Incoming/Outgoing Mail Server set to 172.134.1.20

Note: the Incoming and Outgoing Mail Server fields on the end devices were set directly to the Mail Server's IP address (172.134.1.20) rather than its domain name. This works within a single building, but the e-mail address domain (cse.green.edu.bd) does not match the Mail Server's registered Domain Name (mail_a.green.edu.bd) — a mismatch that is discussed further in Section 12.

11. Connectivity Test (Ping)

Before testing e-mail, basic IP connectivity across the router was verified using the Add Simple PDU tool between devices in Building A and Building B. All ICMP tests completed successfully, as shown in Figure 6, confirming that Router2 correctly routes traffic between the 172.134.1.0/24 and 172.135.1.0/24 subnets.

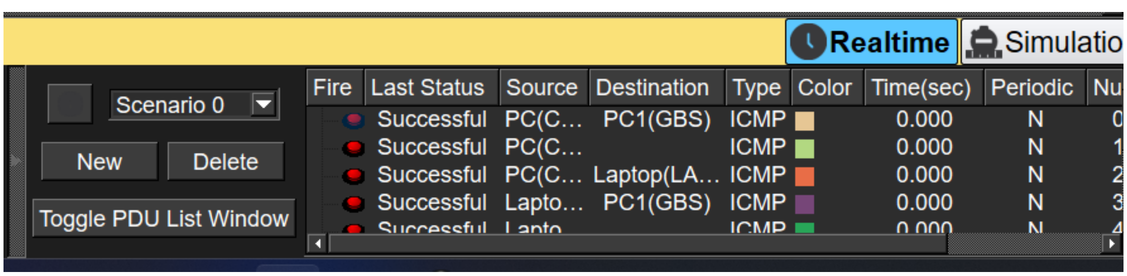

Figure 6: PDU list showing successful ICMP pings between PC(CSE), PC1(GBS), and Laptop(LAW)

12. E-mail Test and Verification

A test e-mail was then composed on PC(CSE) and addressed to chairgbs@gbs.green.edu.bd in Building B. Figure 7 shows the Mail Browser on PC(CSE) before the test, with an empty inbox.

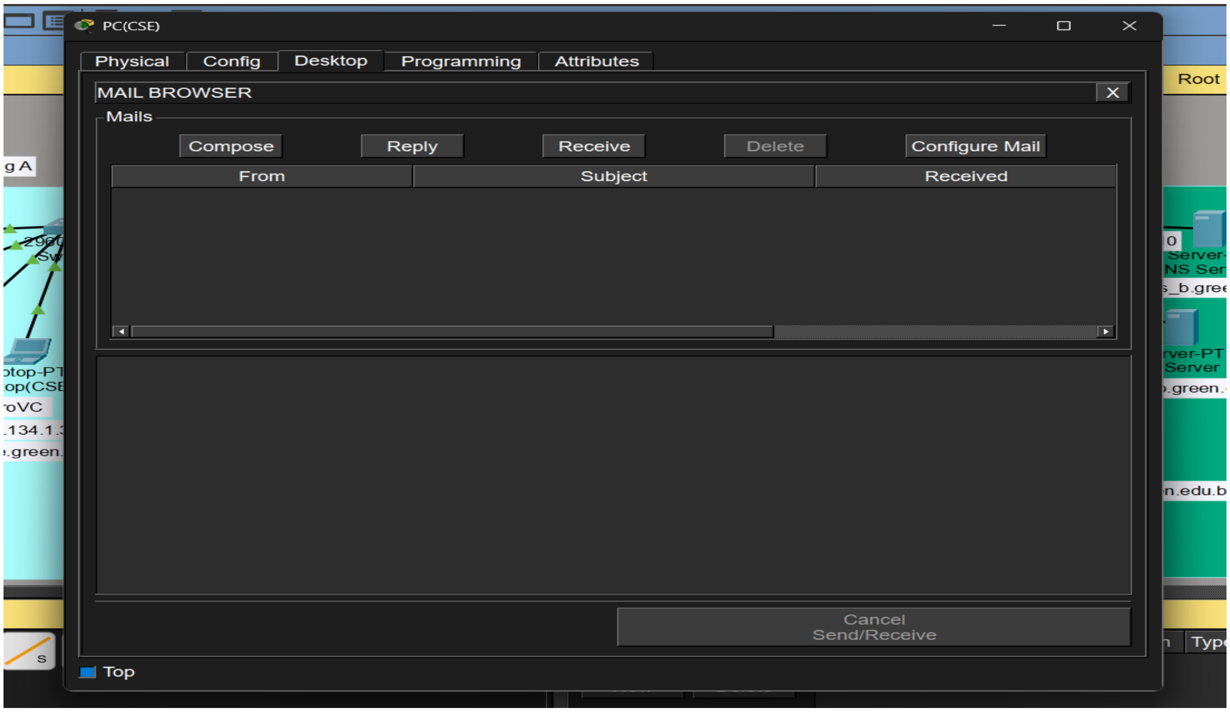

Figure 7: Mail Browser on PC(CSE) before sending the test e-mail

When the message was sent, PC(CSE) reported an SMTP Response Error, as shown in Figure 8.

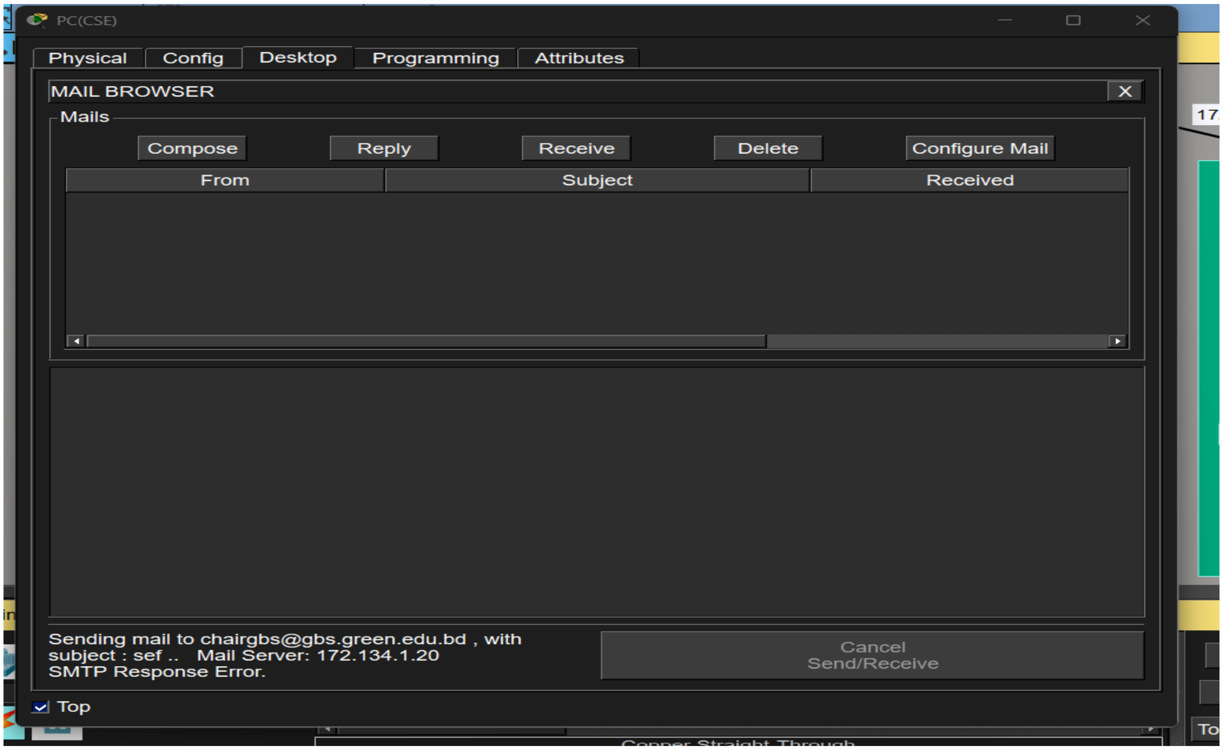

Figure 8: SMTP Response Error returned while sending mail from ChairpersonCSE to chairgbs@gbs.green.edu.bd

On the receiving side, PC1(GBS) was used to check for new mail via the Receive button. The device successfully connected to POP3 Server 172.135.1.20 ("Receive Mail Success"), confirming that the POP3 retrieval mechanism itself works correctly, although no message had arrived because of the SMTP error above (Figure 9).

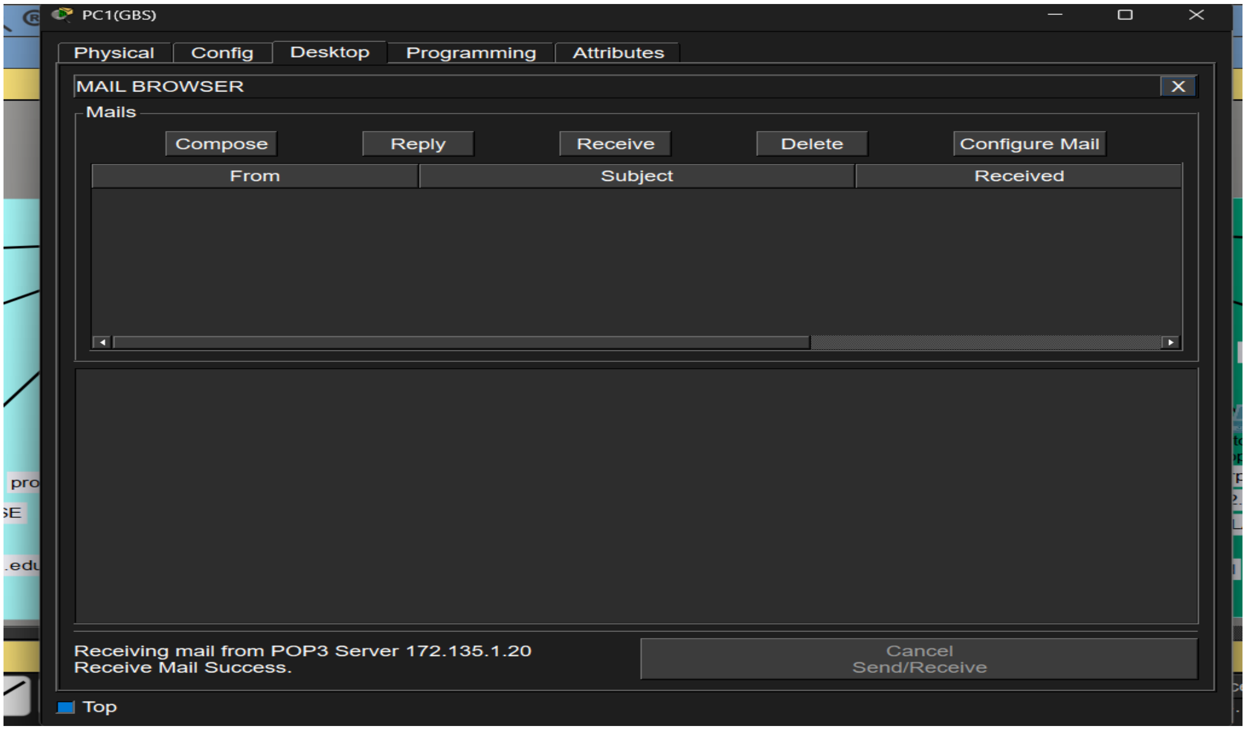

Figure 9: PC1(GBS) successfully connecting to POP3 Server 172.135.1.20 ("Receive Mail Success")

13. Result and Discussion

The ping tests between Building A and Building B devices completed successfully (Figure 6), confirming that the physical topology, IP addressing, and router configuration are all correct and that Layer 3 connectivity exists between the two buildings.

However, sending an e-mail from ChairpersonCSE (Building A) to ChairpersonGBS (Building B) returned an SMTP Response Error (Figure 8), even though the POP3 receive mechanism on the destination device worked correctly (Figure 9). The most likely cause is a domain mismatch: each Mail Server in Packet Tracer is registered with a single Domain Name (mail_a.green.edu.bd and mail_b.green.edu.bd), but the e-mail addresses configured on the end devices use different domains (cse.green.edu.bd, gbs.green.edu.bd, etc.). Packet Tracer's simplified mail server expects a sender/recipient's e-mail domain to match its own registered Domain Name; because it does not, the SMTP transaction is rejected.

13.1 Recommended Fix

- Option 1: Change each end device's e-mail address to use the Mail Server's registered domain, e.g. chaircse@mail_a.green.edu.bd instead of chaircse@cse.green.edu.bd.
- Option 2: Change the Domain Name field on Mail Server A/B to the department domain actually required (e.g., set it to green.edu.bd on both, or configure per-department domains if the Packet Tracer version supports multiple domains per server).
- In addition, ensure the DNS Service toggle on both DNS Server A and DNS Server B is switched to On (Section 8), since it was observed to be Off on DNS Server B at the time of testing.

Once the domain names are aligned and the DNS service is enabled, the same test (ChairpersonCSE → ChairpersonGBS) is expected to complete with the message appearing in ChairpersonGBS's inbox after clicking Receive.

14. Conclusion

In this lab, the Green University of Bangladesh City Campus network was designed and simulated in Cisco Packet Tracer, consisting of two buildings connected through Router2, each with its own switch, DNS Server, and Mail Server. All end devices were assigned static IP addresses, DNS A records were created for each building's mail domain, and e-mail accounts were configured for every user. Connectivity testing (ping) across the router was fully successful. The e-mail test uncovered an SMTP Response Error caused by a domain-name mismatch between the end devices' e-mail addresses and the Mail Servers' registered domain names, which was identified and documented along with the corrective steps needed to achieve successful cross-building e-mail delivery.